U-C-211a

DECEMBER 13, 1956

SUPERSEDING
Int. Fed. Spec. U–C–00211 (GSA–FSS)
January 19, 1956 and
Fed. Spec. U–C–211
March 7, 1940

FEDERAL SPECIFICATION

CEMENT; ZINC PHOSPHATE, DENTAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers zinc-phosphate cements suitable for dental use.

1.2 Classification.

1.2.1 *Type and grade.*—This specification covers but one type and one grade of dental zinc phosphate cement.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

LLL–B–631—Boxes; Fiber, Corrugated (For Domestic Shipment).

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

PPP–B–676 — Boxes; Set-Up, Paperboard.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

American Dental Association:

Specification No. 8—Dental Zinc Phosphate Cement.

(Copies of American Dental Association Specification may be obtained free of charge upon application to the American Dental Association Research Fellowship, National Bureau of Standards, Washington 25, D. C.).

402169/56/No. 178

FSC Class 6520

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:20Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material.—Zinc phosphate cement shall consist of a powder and a liquid which, when mixed as directed, will harden and set to a condition satisfactory for dental use.

3.1.1 *Powder.*—The powder shall be essentially zinc oxide to which may be added one or several modifiers.

3.1.1.1 *Condition.*—The powder shall be free of extraneous materials such as dirt and lint. The pigment shall be uniformly dispersed throughout the powder.

3.1.2 *Liquid.*—The liquid shall be essentially aluminum phosphate and phosphoric acid. Metallic salts may be added as buffers to reduce the rate of reactivity of the liquid with the powder.

3.1.2.1 *Condition.* — The liquid shall be water-clear and there shall be no deposits or sediment on the inside of the glass container.

3.1.2.2 *Volume.*—The liquid shall be supplied in an amount 20 percent in excess of that necessary to combine with the total amount of powder when mixed by testing consistency specified in 4.3.1 or 4.3.2.

3.2 Unset cement.

3.2.1 *Spatulation.*—Cements when spatulated in the usual manner shall not form lumps or granules, or evolve gas.

3.3 Set cement.

3.3.1 *Color.* — Colors for the set cement shall be as specified in the invitation for bids or purchase orders.

3.3.2 *Discoloration.*—The set cement shall not discolor tooth structure.

3.4 Arsenic content.—The cement shall not contain more than one part of arsenic in 500,000 parts of cement (0.0002 percent) when mixed to standard testing consistency.

3.5 Physical properties.

3.5.1 *Time of setting.* — When tested as specified in 4.3.2, the time of setting shall be not less than four minutes nor more than ten minutes.

3.5.2 *Compressive strength.* — The compressive strength shall be not less than 840 kilograms per square centimeter (12,000 pounds per square inch) for specimens crushed 7 days after mixing, when tested as specified in 4.3.3.

3.5.3 *Film thickness.*—The film thickness shall not exceed 0.04 mm. (0.0016-inch) when tested as specified in 4.3.4.

3.5.4 *Solubility and disintegration.*—The disintegration of the cement shall not exceed 0.30 percent by weight after immersion for 7 days in distilled water, when tested as specified in 4.3.5.

3.6 Manufacturer's directions. — Adequate and accurate instructions for proportioning and manipulation shall accompany each unit package. These instructions shall include the required temperature of the slab, the powder-liquid ratio, the rate of incorporating the powder and the time of mixing.

3.7 Lot and control number.—Each container of powder and each container of liquid shall be marked with a serial number or a combination of letters and numbers which shall refer to the manufacturer's records for that particular lot or batch of cement powder or liquid.

3.8 Workmanship.—The powder and liquid shall be uniform and free from poisonous and foreign materials.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Unless otherwise specified, samples shall be submitted in accordance with instructions furnished by the procuring activity. In addition, sampling procedures

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:20Z
Check the source to verify that this is the current version before use.

shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105.

4.2 Inspection.—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at point of delivery designated in the contract or purchase order of procuring agency. In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

4.3 Tests.—Tests shall be conducted in accordance with the applicable specifications, as necessary to determine compliance with this specification.

4.3.1 *Preparation of test specimens.*—The manufacturer or contractors shall furnish the powder-liquid ratio for obtaining the standard testing consistency. (See 6.5.) This ratio shall be used for all test purposes. The preparation of the test specimens shall be conducted at a temperature between 18° and 24°C. (65° and 75°F.) and at a relative humidity between 55 and 75 percent. The mixing technics employed in the preparation of all test specimens shall be those which accompany the packages of cement except that the temperature and humidity may vary within above limits. All apparatus and instruments shall be clean, dry, and free from particles of hardened cement.

4.3.2 *Time of setting.*—A ring of 5 mm. (0.2 inch) high and 10 mm. (0.4 inch) in diameter is placed on a flat plate and filled with cement of standard consistency. Three minutes after starting the mix, the specimen is transferred to an atmosphere of 100 percent relative humidity at 37°C. (99°F.). A standard Gillmore needle weighing 454 grams (1 pound) and having an end 1.06 mm. (0.0417 inch) in diameter shall be lowered vertically until the surface of the cement is touched. This shall be repeated at frequent intervals. The time of setting shall be the number of minutes elapsed from the starting of the mix to the time when the needle fails to make a perceptible circle on the surface of the specimen. The setting time shall be reported to the nearest minute.

4.3.3 *Ultimate compressive strength.*—A cylindrical mold 12 mm. (0.472 inch) high and 6 mm. (0.236 inch) in diameter shall be placed on a flat plate and slightly overfilled with cement of standard consistency. A second flat plate shall be pressed on top of the mold to remove excess cement. The molds shall be made of hard rubber, glass, or other substance that will not be corroded by the cement. (Painting the molds with a thin solution of hard wax in benzene will facilitate removal of the specimens.) Three minutes after starting the mix, the molds shall be transferred to an atmosphere of 100 percent relative humidity at 37°C. (99°F.). Thirty minutes later the specimens shall be immersed in distilled water at room temperature.

4.3.3.1 The ends of the cylinder shall be surfaced plane at right angles to the axis. The ends of the specimens may be ground flat by the use of a small amount of carborundum powder (200 mesh) and water. The molds containing the specimens are drawn back and forth across a glass plate coated with the abrasive and water. They should be rotated about ¼ turn every few strokes. The test specimens shall be kept wet during the grinding and until after crushing. The machine used in crushing the test specimens shall be operated at a rate of speed which will move the crushing head 0.25 mm. (0.01 inch) per minute.

4.3.3.2 The value for compressive strength shall be reported as the average of 3 or more from a lot of 5 specimens and shall be rounded off to the nearest 10 kilograms per square centimeter. If the values for individual speci-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:20Z
Check the source to verify that this is the current version before use.

mens vary more than 15 percent below the average they shall be discarded and the average of the remaining specimens shall be reported. In case more than 2 of the specimens are eliminated, the test shall be repeated.

4.3.4 *Film thickness.*—A portion of a mix of standard consistency is placed between 2 flat square or round plates of uniform thickness. The surfaces of the plates between which the cement is spread shall be approximately 2 square centimeters (0.31 square inch). Three minutes after the mix is started a load of 15 kilograms (33 pounds) shall be applied vertically on the top plate.

4.3.4.1 Ten minutes after the mix is started the thickness of the two plates with the cement between them shall be determined. The difference in the thickness of the plates with and without the cement film shall be considered as the film thickness. An average of three tests shall be reported to the nearest 5 microns (0.0002 inch).

4.3.5 *Disintegration.*—The disintegration of a cement is a measure of the erosion plus the extraction of soluble material from the cement by the action of water.

4.3.5.1 One-half milliliter (0.5 ml.) of cement of standard consistency shall be pressed between two flat plates until the cement disk is 20 mm. (0.79 inch) in diameter (a piece of fine platinum wire placed in the soft cement before the specimens are formed provides a convenient method of holding the specimens). Three minutes after the mix is started the plates and cement shall be placed in an oven at 37°C. (99°F.) in a relative humidity of 100 percent for one hour. Two such specimens shall be used for each determination.

4.3.5.2 After 1 hour the specimens shall be withdrawn from the oven, placed into a tared weighing bottle or flask, and quickly weighed. The combined weight of the two specimens and the weighing bottle, less the weight of the weighing bottle and the platinum wire, shall be taken as the weight of the specimens of cement. The specimens shall be immediately submerged by pouring 50 ml. of distilled water into the weighing bottle, which is stored for 7 days at 37°C. (99°F.). The specimens shall then be removed from the water. There shall be no evidence of crystal growth or extensions from the surface of the specimens. The water shall be evaporated from the flask at a temperature just below 100°C. (212°F.). The flask is then dried at 149°C. (300°F.) to constant weight. The flask and contents are weighed. The difference between the final weight of the flask and its initial weight is the amount of disintegration. The gain in weight divided by the weight of the specimens times 100, shall be the percentage of disintegration. The average of duplicate tests (two flasks containing two specimens each), shall be reported to the nearest 0.1 percent.

4.3.6 *Arsenic test.*—The test for arsenic shall conform to that designated by the American Chemical Society recommended specification for reagent phosphoric acid (see 6.6), and shall be made on samples of cement mixed to standard testing consistency.

5. PREPARATION FOR DELIVERY

(For definitions and levels, see 6.1).

5.1 Packaging.

5.1.1 *Unit of issue.*—One (1) ounce of cement powder, and a sufficient quantity of liquid to provide an amount of 20 percent in excess of that necessary to combine with the specified amount of powder when mixed to standard testing consistency, constitutes one unit.

5.1.2 *Quantities.*—Packaging and packing quantities shall be as specified in table I.

TABLE I.—*Packaging and packing quantities*

Unit pack	Intermediate pack	Shipping container
1 unit	6 units	96 units

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:20Z
Check the source to verify that this is the current version before use.

5.1.2.1 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—The portions of cement and liquid constituting one unit shall be individually packaged in suitable sealed bottles of appropriate size and design. The bottles together with required instructions shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–676. Carton shall contain sufficient cushioning material to prevent shifting and breakage. Carton shall be adequately secured to prevent accidental opening.

5.1.3.2 *Intermediate package.*—The number of unit cartons specified in table I shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification LLL–B–631 or PPP–B–676. Carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Each unit shall be packaged in standard commercial containers of the size and kind commonly used for the purpose, which will afford the degree of protection required for shipment and use of the end product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—The number of units specified in table I shall be overpacked in an exterior container designed for the type 3 load and constructed in accordance with Federal Specification PPP–B–621.

5.2.1.2 *Case liner.*—Each box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.*—Strapping (when required) shall be in accordance with applicable box specification.

NOTE.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—The number of units specified in table I shall be overpacked in an exterior container designed for the type 3 load. Construction, closure, and strapping (when required) shall be in accordance with Federal Specification PPP–B–621.

5.2.3 *Level C minimum military pack.*—The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Bottles.*—Each bottle shall be marked with commercial name, shade, lot (control) number and name of contractor.

5.3.2 *Unit cartons.*—Each unit carton shall be marked with stock number, item identification, unit, quantity, lot (control) numbers, office identifying number, and name of contractor.

5.3.3 *Intermediate cartons and shipping containers.*—Each intermediate carton and shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color marking and symbol are not

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:20Z
Check the source to verify that this is the current version before use.

required on containers for Level A packs when addressed to domestic establishments, for the Medical Services of the Armed Forces.

6. NOTES

6.1 Federal Standard 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 **Ordering data.**—Purchasers should exercise any desired options offered herein. (See 3.3.1, 5.1 and 5.2).

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.5 Reference: Journal, American Dental Association, volume 24, page 2019, December 1937.

6.6 Reference: Industrial and Engineering Chemistry, 19: 1369, December 1927.

6.7 **Transportation description.** — Transportation description applicable to this item is:

Dental Cement.
Carload minimum weight 12,000 pounds, subject to Rule 34, Uniform Freight Classification.

Truckload minimum weight 12,000 pounds.

6.8 This specification covers the following items appearing in the Armed Services Medical Stock List:

6520–514–2350 — Cement, Zinc Phosphate, Dental, Light Gray, 1 oz.

6520–514–2450 — Cement, Zinc Phosphate, Dental, Light Yellow, 1 oz.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

★ U. S. GOVERNMENT PRINTING OFFICE: 1956

Copies of this specification may be purchased for 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:20Z
Check the source to verify that this is the current version before use.